Math 133: Derivative Reference Sheet

$f'(a)$ is the instantaneous rate of change of f and the slope of its tangent line at $(a, f(a))$. Below, c is a constant; all rules apply where the functions involved are defined and differentiable.

Basic derivatives

$$\begin{aligned} \frac{d}{dx}c &= 0 & \frac{d}{dx}x^n &= nx^{n-1} \\ \frac{d}{dx}e^x &= e^x & \frac{d}{dx}\ln x &= \frac{1}{x} \\ \frac{d}{dx}b^x &= b^x \ln b & \frac{d}{dx}\log_b x &= \frac{1}{x\ln b} \end{aligned}$$

For logarithms, $x > 0$; for the base b, assume $b > 0$, $b \neq 1$. Rewrite roots and reciprocals as powers: $\sqrt{x} = x^{1/2}$, $1/x^m = x^{-m}$.

Differentiation rules

Constant multiple	$(cf)' = cf'$
Sum / difference	$(f \pm g)' = f' \pm g'$
Product	$(fg)' = f'g + fg'$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g \neq 0$
Chain	$\frac{d}{dx}f(g(x)) = f'(g(x))\,g'(x)$
Power with chain rule	$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x)$

Tangent line and linear approximation at $x = a$

$$\begin{aligned} \text{Tangent line:} \quad & y - f(a) = f'(a)(x - a). \\ \text{Linear approximation:} \quad & f(x) \approx L(x) = f(a) + f'(a)(x - a) \quad (x \text{ near } a). \end{aligned}$$

Equivalently, $f(a + \Delta x) \approx f(a) + f'(a)\,\Delta x$ for small Δx.

Reading the derivative

$f' > 0$ throughout an interval	f is increasing on that interval.
$f' < 0$ throughout an interval	f is decreasing on that interval.
$f'(a) = 0$	Horizontal tangent; possible local max/min.
f' changes from $+$ to $-$ at a	f has a local maximum at a.
f' changes from $-$ to $+$ at a	f has a local minimum at a.

$f'(a) = 0$ alone does not guarantee a local max/min: $f(x) = x^3$ at $a = 0$. For absolute extrema on a closed interval, also check the endpoints.